Skills Summary

Completing Two-Column Triangle-Congruence Proofs

Use this reference while completing your worksheet or when studying.

How every proof here is built: list what you are given, add the free facts the picture supplies (a shared side or angle, vertical angles, parallel-line angles, midpoints and bisectors), count what kind of pairs you now have — S or A, and in what order — then name the test on the last line. Never assume from appearance.

1. Side-based proofs: SSS and SAS (proof-sss-sas)

The tests: **SSS** — three pairs of sides. **SAS** — two pairs of sides and the angle *between* them.

SSA is not a test. If the angle is not included, the triangles need not be congruent.

Free facts to look for: a shared side ($\overline{AD} \cong \overline{AD}$, Reflexive); a midpoint (gives two congruent segments); vertical angles at an intersection.

Example: E is the midpoint of $\overline{AC}$ and of $\overline{BD}$. Prove $\triangle AEB \cong \triangle CED$.
Step 1. Two midpoints give two pairs of sides, and the angles at E are vertical — sides around an angle points to SAS.
Step 2. Write the pairs in the order S–A–S so the angle sits between the two sides:

Statements	Reasons
1. $\overline{AE} \cong \overline{EC}$	1. Definition of midpoint
2. $\angle AEB \cong \angle CED$	2. Vertical angles are congruent
3. $\overline{BE} \cong \overline{ED}$	3. Definition of midpoint
4. $\triangle AEB \cong \triangle CED$	4. SAS

Try it: $\overline{AB} \cong \overline{AC}$ and D is the midpoint of $\overline{BC}$. Which test proves $\triangle ABD \cong \triangle ACD$, and which pair does the picture hand you for free?

check: SSS — the free pair is $\overline{AD} \cong \overline{AD}$ (Reflexive)

2. Angle-based proofs: ASA, AAS and HL (proof-asa-aas-hl)

The tests: **ASA** — two angles and the side *between* them. **AAS** — two angles and a side *not* between them. **HL** — right triangles only: hypotenuse plus one leg.

ASA and AAS use the same three marks; what decides the name is *where the side sits.*

Example: $\overline{AB} \parallel \overline{DC}$, E is the midpoint of $\overline{AC}$, and B, E, D are collinear. Prove $\triangle ABE \cong \triangle CDE$.

Step 1. Parallel lines give congruent alternate interior angles, the midpoint gives the side between them — angle–side–angle.

Step 2. Check the position of the side before naming the test: $\overline{AE}$ lies between the two marked angles, so it is ASA, not AAS.

Statements	Reasons
1. $\angle BAE \cong \angle DCE$	1. Alternate interior angles
2. $\overline{AE} \cong \overline{CE}$	2. Definition of midpoint
3. $\angle AEB \cong \angle CED$	3. Vertical angles are congruent
4. $\triangle ABE \cong \triangle CDE$	4. ASA

Try it: Two right triangles share the hypotenuse $\overline{BD}$, and one pair of legs is congruent. Which test applies, and why is the similar-looking SSA not a test?

check: HL — the right angle
removes the SSA swing (Pythagoras fixes side 3)

3. After the proof: CPCTC equations (cpctc-algebra)

CPCTC: once two triangles are proved congruent, *Corresponding Parts of Congruent Triangles are Congruent.*

So matching sides are equal in length and matching angles are equal in measure. Set the two expressions equal and solve. CPCTC is a *conclusion* — it can never be the reason two triangles are congruent.

Example: $\triangle ABC \cong \triangle DEF$ with $AB = 4x + 1$ and $DE = 2x + 11$. Find x.

Step 1. $\overline{AB}$ and $\overline{DE}$ correspond (read the vertex order), so CPCTC makes the two expressions equal.

Step 2. $4x + 1 = 2x + 11 \Rightarrow 2x = 10 \Rightarrow \boldsymbol{x = 5}$ (check: both sides give 21)

Try it: $\triangle ABC \cong \triangle DEF$ with $BC = 6x - 5$ and $EF = 4x + 7$. Find x.

check: $\boldsymbol{x = 6}$

(!) Watch out: Match vertices in the order the congruence statement names them — $\triangle WXZ \cong \triangle YZX$ says $W \leftrightarrow Y$, not $W \leftrightarrow Z$. And a shared side or angle is worth a full line of the proof: leaving out the Reflexive step is the most common way a correct-looking proof loses credit.